

Initial Testing Run:

```
MyTrip_Final_Submission ./gradlew test --rerun-tasks

> Task :test
TripControllerTest > getExistingIdReturns200() FAILED
    java.lang.AssertionError at TripControllerTest.java:66
TripControllerTest > postValidTripReturns201() FAILED
    java.lang.AssertionError at TripControllerTest.java:37
TripControllerTest > putValidTripReturns200() FAILED
    java.lang.AssertionError at TripControllerTest.java:82
TripTest > testNegativeFuelPriceThrowsException() FAILED
    java.lang.IllegalArgumentException at TripTest.java:73
TripTest > testNegativeDistanceThrowsException() FAILED
    java.lang.IllegalArgumentException at TripTest.java:66
23 tests completed, 5 failed

> Task :test FAILED

[Incubating] Problems report is available at: file:///Users/angelicaalvarado/Desktop/sites/CS4398/MyTrip_Final_Submission/build/reports/problems/problems-report.html
FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':test'.
> There were failing tests. See the report at: file:///Users/angelicaalvarado/Desktop/sites/CS4398/MyTrip_Final_Submission/build/reports/tests/test/index.html

* Try:
> Run with --scan to get full insights from a Build Scan (powered by Develocity).

Deprecated Gradle features were used in this build, making it incompatible with Gradle 10.
You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.
For more on this, please refer to https://docs.gradle.org/9.5.1/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD FAILED in 43s
4 actionable tasks: 4 executed
MyTrip_Final_Submission open build/reports/tests/test/index.html
```

All Results

Gradle Test Run :test

Gradle Test Run :test

summary

23
tests

5
failures

0
skipped

11.530s
duration

78%
successful

Failed		All					
Child	Name	Tests	Failures	Skipped	Duration	Success rate	
TripControllerTest	com.mytrip.controller.TripControllerTest	8	3	0	8.488s	62%	
TripTest	com.mytrip.model.TripTest	6	2	0	0.053s	66%	

Figure 1. Initial failed tests

Visible test that failed and success rates of test.

Second round of testing:

All Results

⊗ Gradle Test Run :test

Gradle Test Run :test

summary

24
tests

6
failures

0
skipped

9.412s
duration

75%
successful

Failed

All

Child	Name	Tests	Failures	Skipped	Duration	Success rate
TripTest	com.mytrip.model.TripTest	6	3	0	0.037s	50%
MytripApplicationTests	com.mytrip.MytripApplicationTests	1	1	0	4.433s	0%
TripServiceTest	com.mytrip.service.TripServiceTest	9	2	0	0.288s	77%

Figure 2. Intermediate JUnit test run showing failures discovered during testing.

Trip Test: Expected java.lang.IllegalArgumentException to be thrown, but nothing was thrown.

The initial automated test run exposed problems in the model, service, and application-context tests. These failures were reviewed to determine whether they resulted from incorrect test assumptions or missing validation in the production code.

Third round of testing:

All Results

● Gradle Test Run :test

Gradle Test Run :test

summary

24	0	0	10.406s	100%
tests	failures	skipped	duration	successful

Child	Name	Tests	Failures	Skipped	Duration	Success rate
TripControllerTest	com.mytrip.controller.TripControllerTest	8	0	0	2.042s	100%
TripTest	com.mytrip.model.TripTest	6	0	0	0.058s	100%
MytripApplicationTests	com.mytrip.MytripApplicationTests	1	0	0	6.570s	100%
TripServiceTest	com.mytrip.service.TripServiceTest	9	0	0	0.411s	100%

all > com.mytrip.controller.TripControllerTest

● Gradle Test Run :test

TripControllerTest

summary standard output

8	0	0	2.042s	100%
tests	failures	skipped	duration	successful

Child	Tests	Failures	Skipped	Duration	Success rate
deleteExistingTripReturns204()	1	0	0	0.018s	100%
getAllTripsReturns200()	1	0	0	0.029s	100%
getExistingIdReturns200()	1	0	0	0.609s	100%
getMissingIdReturns404()	1	0	0	0.021s	100%
getOwnerSpecificTripsReturns200()	1	0	0	0.026s	100%
postInvalidTripReturns400()	1	0	0	0.246s	100%
postValidTripReturns201()	1	0	0	0.046s	100%
putValidTripReturns200()	1	0	0	0.017s	100%

all > com.mytrip.model.TripTest

● Gradle Test Run :test

TripTest

summary

6	0	0	0.058s	100%
tests	failures	skipped	duration	successful

Child	Tests	Failures	Skipped	Duration	Success rate
testExactlyOnBudgetTrip()	1	0	0	0s	100%
testNegativeDistanceThrowsException()	1	0	0	0.001s	100%
testNegativeFuelPriceThrowsException()	1	0	0	0.037s	100%
testOverBudgetTrip()	1	0	0	0.003s	100%
testStandardTripCalculations()	1	0	0	0.011s	100%
testZeroMpgThrowsException()	1	0	0	0.001s	100%

all > com.mytrip.MytripApplicationTests

● Gradle Test Run :test

MytripApplicationTests

summary standard output

1	0	0	6.570s	100%
tests	failures	skipped	duration	successful

Child	Tests	Failures	Skipped	Duration	Success rate
contextLoads()	1	0	0	0.398s	100%

all > com.mytrip.service.TripServiceTest

● Gradle Test Run :test

TripServiceTest

summary

9	0	0	0.411s	100%
tests	failures	skipped	duration	successful

Child	Tests	Failures	Skipped	Duration	Success rate
createTripCallsRepositorySave()	1	0	0	0.003s	100%
deleteTripDeletesExistingTrip()	1	0	0	0.379s	100%
deleteTripThrowsWhenMissing()	1	0	0	0.003s	100%
getAllTripsReturnsRepositoryResults()	1	0	0	0.003s	100%
getTripByIdReturnsExistingTrip()	1	0	0	0.002s	100%
getTripByIdThrowsWhenMissing()	1	0	0	0.002s	100%
getTripsForOwnerUsesOwnerEmailQuery()	1	0	0	0.003s	100%
nullTripsRejectedOnCreate()	1	0	0	0.002s	100%
updateTripChangesValuesAndCallsSave()	1	0	0	0.007s	100%

Figure 2. Final Gradle test report showing 24 tests, 0 failures, and 100% success.

After correcting the test setup and adding appropriate model validation, the complete Gradle test suite was executed again. All 24 automated tests passed with no failures or skipped tests.